

AKIAK, AK, USA

WMO: 702686

Lat: 60.903N

Lon: 161.231W

Elev: 9

StdP: 101.22

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00106

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.8	-27.8	-34.1	0.2	-30.3	-31.3	0.2	-27.6	18.4	-13.2	16.6	-12.4	2.6	30	0.926

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.0	18.3	14.3	16.8	13.3	15.2	12.8	15.0	17.3	14.0	15.9	13.1	14.8	5.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	9.9	15.8	13.0	9.3	14.8	12.2	8.9	14.1	41.9	17.3	39.2	15.9	37.0	14.8	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.6	14.1	12.3	DB	-30.9	23.1	2.8	3.3	-32.9	25.4	-34.6	27.3	-36.2	29.2	-38.2	31.5	(4)
			WB	-30.3	16.8	3.5	2.4	-32.8	18.6	-34.8	20.0	-36.7	21.3	-39.3	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.8	-13.3	-10.6	-11.6	-4.5	3.2	8.9	11.2	11.5	8.7	1.1	-5.6	-9.3	(6)
		DBStd	10.81	9.44	8.29	7.28	5.84	4.41	3.15	2.86	2.13	2.79	5.57	6.64	7.46	(7)
		HDD10.0	4100	723	576	669	437	214	58	18	52	276	469	599		(8)
		HDD18.3	6991	981	809	927	687	470	284	222	212	288	534	719	857	(9)
		CDD10.0	152	0	0	0	0	3	24	56	55	14	0	0	0	(10)
		CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
		CDH23.3	8	0	0	0	0	0	0	8	0	0	0	0	0	(12)
		CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.8	5.8	6.5	6.7	5.9	5.1	5.3	5.1	5.3	5.6	6.0	6.3	6.1	(14)
(15)	Precipitation	PrecAvg	488	22	21	19	19	28	43	64	81	75	44	44	29	(15)
(16)		PrecMax	600	65	77	91	54	68	110	99	161	176	109	105	86	(16)
(17)		PrecMin	317	2	1	2	1	3	7	18	29	25	4	0	6	(17)
(18)		PrecStd	73	16	16	17	13	17	19	24	29	35	25	26	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	2.7	1.9	2.2	8.6	16.2	19.6	23.2	19.0	16.2	11.0	5.5	3.6		(19)
(20)		0.4% MCWB	1.5	0.9	1.4	5.1	11.1	14.6	17.3	14.5	12.6	9.6	4.6	2.6		(20)
(21)		2% DB	1.5	1.3	1.2	5.7	13.0	17.2	19.1	17.4	14.1	9.6	4.5	2.4		(21)
(22)		2% MCWB	0.7	0.7	0.7	3.8	9.7	13.1	15.6	13.4	11.9	8.5	3.9	1.7		(22)
(23)		5% DB	0.8	0.9	0.4	3.4	11.3	15.2	17.1	16.0	13.0	8.5	3.4	1.4		(23)
(24)		5% MCWB	0.4	0.5	0.0	2.2	8.6	12.2	14.0	13.2	11.3	7.5	2.8	0.8		(24)
(25)		10% DB	-0.2	0.2	-0.9	2.0	8.9	13.4	15.1	14.6	12.3	7.4	2.5	0.8		(25)
(26)		10% MCWB	-0.5	-0.2	-1.2	1.0	7.2	11.1	13.1	12.9	10.8	6.5	1.8	0.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	1.7	1.3	1.6	5.7	11.7	15.0	18.9	15.5	13.9	10.1	5.0	3.0		(27)
(28)		0.4% MCDB	2.3	1.5	2.0	8.3	14.7	18.7	22.6	17.7	15.3	10.9	5.2	3.2		(28)
(29)		2% WB	1.0	0.8	0.8	3.8	10.1	13.7	15.8	14.5	12.5	8.8	3.9	1.7		(29)
(30)		2% MCDB	1.3	1.0	1.1	5.6	13.0	16.5	18.2	16.1	13.7	9.5	4.4	2.1		(30)
(31)		5% WB	0.3	0.4	0.0	2.3	8.7	12.4	14.4	13.7	11.7	7.8	2.9	0.9		(31)
(32)		5% MCDB	0.7	0.7	0.3	3.4	10.8	14.6	16.6	15.1	12.8	8.4	3.3	1.3		(32)
(33)		10% WB	-0.7	-0.2	-1.1	1.1	7.6	11.4	13.2	13.0	10.8	6.7	2.0	0.2		(33)
(34)		10% MCDB	-0.3	0.1	-0.8	1.8	9.1	13.3	15.0	14.4	11.9	7.4	2.4	0.6		(34)
(35)	Mean Daily Temperature Range	MDBR	5.6	6.6	6.9	6.3	5.4	4.8	4.0	4.0	4.0	4.3	5.0	5.7		(35)
(36)		5% DB MCDBR	5.5	4.1	5.5	6.0	8.8	8.4	7.8	6.2	5.2	3.4	3.6	4.3		(36)
(37)		5% WB MCWBR	5.3	3.9	5.2	4.5	5.7	5.1	4.3	3.3	3.3	3.0	3.7	4.0		(37)
(38)		5% WB MCDBR	5.4	4.4	5.2	5.8	8.4	7.4	6.9	5.0	4.2	3.1	3.6	4.5		(38)
(39)		5% WB MCWBR	5.2	4.3	5.0	4.4	5.5	4.8	4.1	3.1	3.1	3.0	3.7	4.3		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.228	0.283	0.290	0.349	0.380	0.366	0.370	0.382	0.342	0.296	0.247	0.215		(40)
(41)		taud	2.084	2.172	2.234	2.181	2.280	2.393	2.421	2.374	2.447	2.415	2.076	1.926		(41)
(42)		Ebn at Noon	518	695	834	838	841	862	848	803	776	694	464	327		(42)
(43)		Edh at Noon	39	70	92	118	117	107	101	98	77	55	39	28		(43)
(44)	All-Sky Solar Radiation	RadAvg	0.37	1.06	2.92	4.51	5.08	5.22	4.32	3.36	2.37	1.20	0.43	0.18		(44)
(45)		RadStd	0.06	0.14	0.32	0.40	0.48	0.42	0.42	0.37	0.28	0.09	0.07	0.02		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	+1.67	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page